

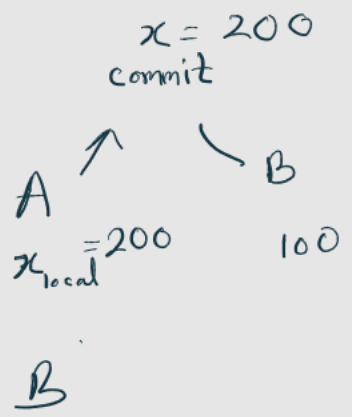
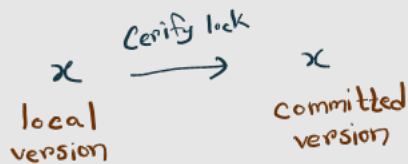
Concurrency Control

↳ Isolation maintain

- Type :
1. Multiversion Timestamp Ordering (MVTO)
 2. Multiversion Two-Phase Locking Using Certify Lock (MV2PL) (C-Lock)
 3. Two Phase Lock (2PL)
 4. Graph Based Concurrency control
- Multiversion Concurrency control



2. Multiversion Two Phase Lock



- Read lock
- Write lock
- Certify lock

A

Read
Write
Certify

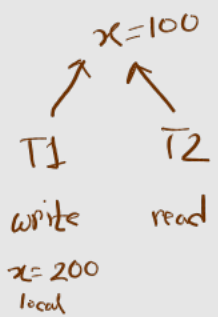
Read	Write	Certify
✓	✓	✗
✓	✗	✗
✗	✗	✗

- ① In a company database, Transaction T1 updates an employee's salary and creates a new version of the salary. Before T1 commits, Transaction T2 wants to read the same employee's salary. Instead of waiting for T1 to finish, T2 reads the previously committed version.

Answer:

The concurrency control method being used is Multiversion Two-Phase Locking (MV2PL).

T2 does not need to wait because multiple versions of the same data are maintained. Since T1's new version has not been committed yet, T2 can read the previously committed version.



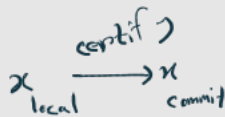
- ② In a banking database, T1 updates Account A and creates a new version called A1. Before committing A1, T1 must obtain a special lock. Meanwhile, T2 wants to read Account A while T1 is still updating it.

Questions:

1. What special lock does T1 need before committing A1?
2. Can T2 continue reading without waiting for T1?
3. Which version will T2 read?

Answer:

1. T1 must obtain a **Certify Lock (C-lock)** before committing A1.
2. Yes, T2 can continue without waiting.
3. T2 reads the ^{previous} ~~latest~~ committed version of Account A. Since A1 is not committed yet, T2 reads the previous committed version.



1. Multiversion Timestamp Ordering (MVTO)

$x \rightarrow x_0, x_1, x_2, \dots$

$x_0 \rightarrow \text{Write_TS}(x_0), \text{Read_TS}(x_0)$

$\text{TS}(T1) = 5$

$\text{TS}(T2) = 10$

$T1 \rightarrow$	$T2 \rightarrow$
<u>5</u>	10
$x_0 =$	RTS
	<u>5</u>
	WTS

Rules :

* Select version/READ

$\text{TS}(T_i) \geq \text{Write_TS}(x_i)$ careful (1)

[Take maximum]

(careful (2))

* WRITE

(i) $\text{TS}(T_i) < \text{Read_TS}(x_i)$ Abort/Rollback

(ii) $\text{TS}(T_i) = \text{Write_TS}(x_i)$ Overwrite

(iii) $\text{TS}(T_i) > \text{Write_TS}(x_i)$ New version

$x_0 \rightarrow x_1$

$x_1 \rightarrow x_2$

Example :

① $\text{TS}(T1) = 7$

	<u>RTS</u>	<u>WTS</u>
x_0	5	0
x_1	10 10	5 ✓
x_2	15	10

Read(x_i)

Ans :

x_1 ; $\text{RTS}(x_1) = \max(7, 10) = 10$

Select
version

RTS/WTS update

② $\text{TS}(T1) = 10$

$\text{WTS}(x_1) = 5$

$\text{RTS}(x_1) = 5$

$\text{WTS}(x_2) = 11$

$\text{RTS}(x_2) = 12$

Write(x_i)

<u>Ans :</u>	RTS	WTS
$\rightarrow x_1$	5	5
x_2	12	11
x_3		10

x_3 ; $\text{WTS}(x_3) = 10$

$\text{TS}(T1) > \text{Write_TS}(x_i)$ New version
 $10 > 5$

③ $TS(T1) = 5$
 $TS(T2) = 10$

$T1 \Rightarrow Read(x_i)$
 $T1 \Rightarrow Write(x_i)$
 $T2 \Rightarrow Read(x_i)$
 $T2 \Rightarrow Write(x_i)$
 $T1 \Rightarrow Read(x_i)$
 $T1 \Rightarrow Write(x_i)$

(a) Complete
(b) x এর গুলোয়
WTS RTS

Ans:

(a)

$T1 \Rightarrow Read(x_i)$ 5	$x_0 ; RTS(x_0) = \max(5, 0) = 5$
$T1 \Rightarrow Write(x_i)$ 5	$x_1 ; WTS(x_1) = 5$
$T2 \Rightarrow Read(x_i)$ 10	$x_1 ; RTS(x_1) = \max(0, 10) = 10$
$T2 \Rightarrow Write(x_i)$ 10	$x_2 ; WTS(x_2) = 10$
$T1 \Rightarrow Read(x_i)$ 5	$x_1 ; RTS(x_1) = \max(5, 10) = 10$ <i>careful</i>
$T1 \Rightarrow Write(x_i)$ 5	$x_1 ; ROLLBACK$

(b)

	<u>RTS</u>	<u>WTS</u>
x_0	5	0
x_1	10	5
x_2	-	10

4

Ans:

Time	10:00	10:21	10:33
Transaction	T1	T2	T3
X	10	11	12

$TS(T1) = 10$
 $TS(T2) = 11$
 $TS(T3) = 12$

T1	T2	T3
R(Xi)	W(Xi)	
	R(Xi)	W(Xi)
W(Xi)		R(Xi)

read_TS((Xi) =
write_TS(Xi) =

	<u>RTS</u>	<u>WTS</u>
x_0	0 10	0
x_1	11 12	11
x_2		12

$T1 \Rightarrow Read(x_i)$
10

$x_0 ; RTS(x_0) = \max(0, 10) = 10$

$T2 \Rightarrow W(x_i)$
11

$x_1 ; WTS(x_1) = 11$

$T2 \Rightarrow R(x_i)$
11

$x_1 ; RTS(x_1) = \max(11, 11) = 11$

$T3 \Rightarrow W(x_i)$
12

$x_2 ; WTS(x_2) = 12$

$T1 \Rightarrow W(x_i)$
10

$x_0 ; WTS(x_0) = 10$ [overwrite]

$T3 \Rightarrow R(x_i)$
12

$x_1 ; RTS(x_1) = 12$

$TS(T_i) < Read_TS(x_i)$ Abort/Rollback

$TS(T_i) = Write_TS(x_i)$ Overwrite

$TS(T_i) > Write_TS(x_i)$ New version

The operations are executed in the following order:

Step	Operation	Result(Version Read/Created, TS Updates, Abort?)
1	T1: Read(X)	
2	T2: Read(X)	
3	T1: Write(X)	
4	T3: Read(X)	
5	T2: Write(X)	
6	T3: Write(X)	
7	T1: Read(X)	
8	T2: Read(X)	

	RTS	WTS
x_0	0 12	0
x_{12}	12	12

- $T1 : \text{Read}(x)$
[3]
 $x_0 ; \text{RTS}(x_0) = 3$
- $T2 : \text{Read}(x)$
[7]
 $x_0 ; \text{RTS}(x_0) = 7$
- $T1 : \text{Write}(x)$
[3]
Abort $T1$ [$\text{TS}(T1) < \text{RTS}(x_0)$]
- $T3 : \text{Read}(x)$
[12]
 $x_0 ; \text{RTS}(x_0) = 12$
- $T2 : \text{Write}(x)$
[7]
Abort $T2$ [$\text{TS}(T2) < \text{RTS}(x_0)$]
- $T3 : \text{Write}(x)$
[12]
 $x_{12} ; \text{WTS}(x_{12}) = 12$
- $T1 : \text{Read}(x)$
[3]
skipped [$T1$ already aborted]
- $T2 : \text{Read}(x)$
[7]
skipped [$T2$ already aborted]